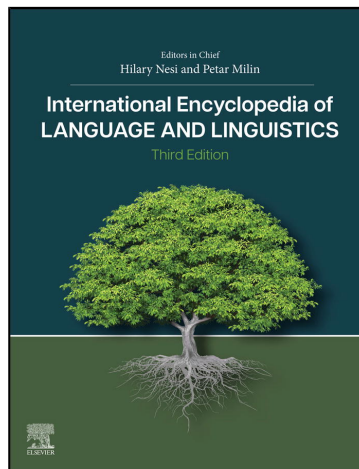


Provided for non-commercial research and educational use.
Not for reproduction, distribution or commercial use.

This chapter was originally published in *International Encyclopedia of Language and Linguistics*, 3e (LAL3), published by Elsevier, and the attached copy is provided by Elsevier for the author's benefit and for the benefit of the author's institution, for non-commercial research and educational use, including without limitation, use in instruction at your institution, sending it to specific colleagues who you know, and providing a copy to your institution's administrator.



All other uses, reproduction and distribution, including without limitation, commercial reprints, selling or licensing copies or access, or posting on open internet sites, your personal or institution's website or repository, are prohibited. For exceptions, permission may be sought for such use through Elsevier's permissions site at:

<https://www.elsevier.com/about/policies/copyright/permissions>

Duñabeitia, J. A., & Pliatsikas, C. (2026). The Multilingual Picture Database (MultiPic). In: Nesi, H., Milin, P. (Eds.), *International Encyclopedia of Language and Linguistics*, 3e. Vol 12., pp. 596–600. UK: Elsevier.

<https://dx.doi.org/10.1016/B978-0-323-95504-1.00023-5>.

ISBN: 9780323955041

Copyright © 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

The Multilingual Picture Database (MultiPic)

Jon Andoni Duñabeitia^a and Christos Piatsikas^b, ^aUniversidad Nebrija, Madrid, Spain; and ^bSchool of Psychology and Clinical Language Sciences, University of Reading, Reading, United Kingdom

© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Introduction	596
The Genesis of MultiPic	597
MultiPic's Global Expansion	597
MultiPic and Linguistic Diversity	597
Expanding the Scope: The Future of MultiPic	598
Conclusion	599
References	599

Key Points

- We review MultiPic, a tool offering linguistic norms for 500 images across 30+ languages to support psycholinguistics research.
- We address the need for culturally diverse research tools and outlines the global collaboration behind MultiPic's development.
- The significance of MultiPic in promoting linguistic diversity by including underrepresented languages and dialects is highlighted.
- The chapter envisions expanding MultiPic to further enhance its utility in research by covering more minority and endangered languages.

Abstract

This chapter reviews Multipic, a research tool comprising linguistic norms for 500 colored object images in over 30 languages or language varieties. As documented by its increasing popularity, Mutlipic has rapidly become an invaluable resource for experts in the field of psycholinguistics, not only because it provides norms in many languages, including underrepresented ones, but also a large set of translation equivalents across the same languages.

Introduction

In the fields of cognitive science and psycholinguistics, the quest for resources that can transcend cultural and linguistic barriers is still ongoing. Normative sets of images and drawings have become indispensable in research across disciplines, acting as standardized tools to explore language, perception, memory, and attention in diverse linguistic groups. This pursuit of creating universally relevant and comprehensive datasets has navigated through various challenges and significant developments.

One of the earliest and most widely recognized datasets in this field is the Snodgrass and Vanderwart picture set, created in 1980 (Snodgrass & Vanderwart, 1980). This collection, comprising black-and-white line drawings, was initially normed in American English but later adapted for various languages, including British English, Chinese, Dutch, French, Italian, Japanese, Spanish and Greek, among others (e.g., Dimitropoulou et al., 2009). Despite its broad usage, this dataset has its limitations. The black-and-white line drawings were found to yield weaker recognition (namely, lower naming accuracies and longer response times) compared to colored pictures, prompting a shift toward developing colored image datasets in different languages, as seen in the work of Rossion and Pourtois (2004). Another significant constraint of the Snodgrass and Vanderwart set was its size of only 260 drawings, which became a serious limitation in experimental design. This led to the creation of new drawings and the norming of these in various languages as part of projects like the International Picture-Naming Project (IPNP). However, the IPNP dataset, which included 795 black-and-white drawings from multiple sources, suffered from heterogeneity in pictorial styles and graphic properties, reducing its efficacy as a universally applicable tool.

In response to these limitations, more extensive and diverse datasets were developed. The Bank of Standardized Stimuli (BOSS; Brodeur et al., 2010) includes photos of nearly 1500 pictures with various norms. However, the BOSS dataset is predominantly limited to American English, with a subset for Canadian French, with norms in a few more languages only recently emerging through various independent investigations (e.g. Brazilian Portuguese, Thai, Russian, Dutch; e.g., Santos et al., 2019). Notably,

the available norming data in these languages vary substantially in many ways, including in terms of the number of items that were normed, and the number of norming dimensions. This highlights the ongoing challenge of developing resources that are truly comprehensive in linguistic scope, including providing consistent norms across a wide variety of languages.

In summary, the evolution of normative sets of images and drawings in scientific research has been driven by the need for more extensive, diverse, and culturally inclusive datasets. While significant progress has been made, as evidenced by projects like BOSS, challenges remain in creating tools that can comprehensively cover the vast array of human languages and cultures. Large-scale studies and collaborations appear to be the most effective way forward, offering a more universal approach to cognitive and linguistic research across different populations. The Multilingual Picture Database (MultiPic) was proposed as a pioneering initiative in this domain, offering a normed dataset of images enriched with linguistic data across many languages, as well as of different varieties of largely spoken languages (currently, varieties of English, Dutch, French, Greek and Spanish). This chapter explores the genesis, evolution, and profound impact of MultiPic, highlighting its role in fostering global research collaboration.

The Genesis of MultiPic

The inception of MultiPic marked a significant milestone in cognitive research. In a seminal study by Duñabeitia et al. (2018), the team presented a well-curated collection of 750 colored object drawings, each standardized and normed across six European languages: British English, Spanish, French, Italian, German and Dutch (the latter separately for the varieties spoken in the Netherlands and in Belgium). This effort was not just about creating a database of pictures; it was an ambitious attempt to embed these images with rich linguistic data, including name agreement, familiarity, visual complexity, and preferred names. This initial step laid a robust foundation for the tool's future expansions and its acceptance in the research community.

Since its launch in April 2018, the first version of MultiPic has achieved remarkable recognition in the research community. By the end of 2023, according to Google Scholar, it had been cited 240 times across various fields, including Psychology, Arts and Humanities, Social Sciences, Neuroscience, Biochemistry, Medicine, and Computer Sciences. This interdisciplinary reach is a testament to the versatility and relevance of MultiPic in advancing cognitive research. The tool has not only facilitated novel research methodologies but has also inspired new lines of inquiry across diverse domains.

MultiPic's Global Expansion

Following its initial success, the MultiPic project subsequently underwent a significant global expansion, as detailed in Duñabeitia et al. (2022). 500 colored pictures of the original database, chosen because of their high naming agreement across the original languages, underwent a massive normative process, resulting in MultiPic currently offering naming norms and familiarity scores across an extensive range of thirty-three languages and language varieties worldwide. Such an expansion not only broadened the geographical reach of the project but also significantly enhanced its linguistic diversity. This transformation of MultiPic into a global tool opened new possibilities for comparative cognitive research, accommodating a wide spectrum of cultural and linguistic backgrounds. The full set of norms in all published and forthcoming languages can be found here: <https://doi.org/10.6084/m9.figshare.19328939.v5>.

The study underscored the importance of having a diverse and comprehensive set of images for research, especially in a world where multilingualism is increasingly common. This expansion addressed previous limitations found in other datasets, such as their restricted linguistic diversity and limited number of images and. By employing a consistent methodology across different languages and cultures, MultiPic provided a unique opportunity for researchers to conduct more inclusive and representative studies in cognitive science.

Furthermore, the study emphasized the value of this expansion in facilitating research in various fields of psychology, cognitive neuroscience, and psycholinguistics, especially in cross-linguistic contexts. The inclusion of a diverse range of languages, from widely spoken ones like British English and Cantonese (Zhong et al., 2024) to less commonly studied languages, like Hungarian or Slovak, and language varieties, like Malay English or Lebanese Arabic, marked a significant step toward a more inclusive approach in cognitive and linguistic research. The expansion of MultiPic thus not only enriched the tool itself but also has the potential of contributing to a broader understanding of cognitive processes across different linguistic and cultural groups. Overall, the global expansion of MultiPic has represented a critical advancement in the field, enabling more comprehensive and culturally sensitive research in cognitive science and related disciplines.

MultiPic and Linguistic Diversity

MultiPic's impact extends beyond just being a research tool; it is a catalyst for promoting inclusivity in cognitive research. The tool supports the process of generating norms in minority and regional languages, as well as languages from the Global South; indeed, at the time of writing of this chapter, MultiPic is being normed in languages such as Albanian, Vietnamese, Brazilian Portuguese and Galician, to name a few, which will soon be made available through future publications. This initiative not only broadens the scope of research but also brings to the forefront the linguistic and cultural richness of communities that are often underrepresented in

cognitive science literature. This makes MultiPic particularly useful for researchers seeking to understand how perception and language processing may vary across different linguistic and cultural groups. By offering a uniform set of stimuli normed across a variety of languages, MultiPic facilitates a more nuanced understanding of these processes. This becomes particularly important when studying different varieties of the same language, where it is possible that the same images may yield different outcomes, due to cultural, linguistic or other factors.

For those researching multilingualism, MultiPic is an invaluable repository of data. It provides a comprehensive set of translation equivalents associated with various concepts across several languages and language varieties. This aspect of MultiPic saves significant time and effort for researchers that would otherwise have to spend it in norming specific word pairs in various language combinations (see [Thompson et al., 2020](#), for a large-scale study exploring semantic alignment of translation equivalents across languages). To exemplify this, consider the series of concepts illustrated in [Fig. 1](#) (namely, a robe, a couch, a trash can and a ruler, according to their naming norms in American English). These seemingly simple and common entities are high-familiarity elements across cultures, and most but not all show remarkable reliability in their name agreement scores across different languages and language varieties, making them excellent examples to demonstrate the value of MultiPic for studies in different linguistic contexts. At the same time, these examples demonstrate the value of norming in different varieties of the same language, as some images may yield the same name across all varieties, but some others may not (see [Table 1](#)). This highlights an important caveat in using norms of a standard language to speakers of a variety of the same language (e.g., using norms in Standard Greek to speakers of Cypriot Greek). These examples, while illustrative, underline the potential of MultiPic in providing concrete, quantifiable data that can be used to compare and contrast cognitive processing across languages in a variety of linguistic and cultural contexts. Moreover, the importance of such a dataset in clinical research and practice cannot be understated (see [Obidziński & Nieznański, 2022](#), for a study with dyslexic adults; see also [Peristeri et al., 2023](#), for a study on autism). This may result especially useful for clinicians working with multilingual patients, as this tool can help assess their linguistic abilities and knowledge in the different languages known to them but on the exact same items.

Expanding the Scope: The Future of MultiPic

One of the most significant aspects of MultiPic is its open-source nature. By making this resource freely available to the research community, it encourages collaboration and knowledge sharing on a global scale. This approach democratizes access to high-quality research materials and is particularly beneficial for researchers in regions with limited resources. The open-source nature of MultiPic not only fosters a spirit of collective knowledge-building but also ensures that the tool continues to evolve and improve through community contributions.

The ongoing development of MultiPic heralds an exciting phase in cognitive research. This database is evolving beyond its initial scope as an image database into a versatile tool tailored for a wide range of linguistic and cognitive studies. Currently ongoing and future expansions are poised to include a diverse array of languages, with particular emphasis on those from lesser-represented regions. This includes minority and/or endangered languages (e.g. Maltese, Scottish Gaelic) that often lack such psycholinguistics tools, meaning that MultiPic may have a role in promoting and reviving such languages and the clinical and experimental work done with them as vehicular languages. This expansion will not only enrich the dataset but also provide invaluable resources for studies that explore linguistic diversity and the nuances of language-mediated conceptualization across different cultures and communities.

The success of MultiPic is not just a testament to the efforts of its creators but also to the enthusiasm of the global research community that has embraced and contributed to it. Moving forward, fostering a more collaborative environment where researchers can share their findings, methodologies, and insights using MultiPic will be vital. The tool is being currently used in hundreds of ongoing research actions across the world, and it is becoming common to see images of MultiPic being used in presentations at conferences, workshops, or online forums dedicated to research dissemination. The use of MultiPic as test material in published research articles is increasingly observed across various domains. This trend is evident in clinical trials, such as [Kuhne et al.'s \(2023\)](#) examination of the impact of cardiovascular exercise on learning, as well as in neuroscientific studies on language production and perception, like the study on lexico-semantic competition by. Additionally, less language-centered investigations like those

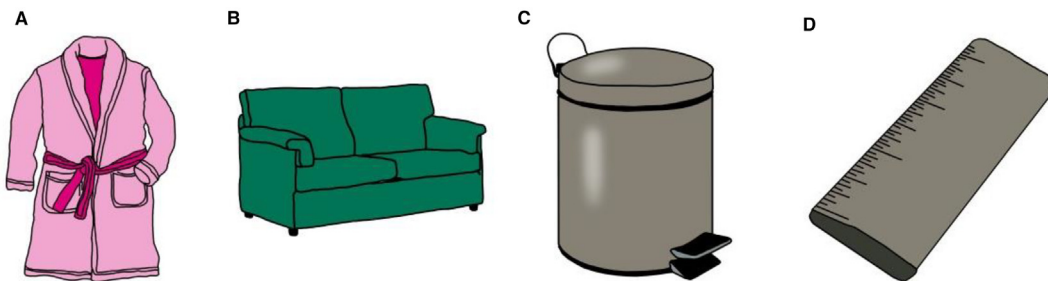


Fig. 1 Examples of four pictorial elements from the MultiPic database.

Table 1 Modal responses for each of the elements depicted in Fig. 1 across languages and varieties.

Languages or varieties	Modal responses			
	A	B	C	D
Arabic (Lebanese)	روب	كنبایه	زباله	مسطرة
Basque	bata	sofa	zakarrontzi	erregela
Catalan	barnús	sofà	paperera	regla
Chinese				
Cantonese	浴袍	沙發	垃圾桶	尺
Mandarin	睡衣	沙发	垃圾桶	尺子
Czech	župan	gauč	koš	pravitko
Dutch				
Standard	badjas	bank	prullenbak	liniaal
Flemish	badjas	zetel	vuilbak	meetlat
English				
American	robe	couch	trash can	ruler
Australian	robe	couch	bin	ruler
British	dressing gown	sofa	bin	ruler
Malay	bathrobe	sofa	bin	ruler
Finnish	kylpytakki	sohva	roskakori	viivotin
French				
Metropolitan	peignoir	canapé	poubelle	règle
Québécois	robe de chambre	divan	poubelle	règle
German	bademantel	sofa	muelleimer	lineal
Greek				
Standard	Ρόμπα	καναπές	κάδος	χάρακας
Cypriot	Ρόμπα	καναπές	κάλαθος	ρίγα
Hebrew	חלוק	ספה	נפ	סרגל
Hungarian	Köntös	kanapé	kuka	vonalzó
Italian	accappatoio	divano	cestino	righetto
Korean	가운	소파	쓰레기통	자
Malay	baju mandi	sofa	tong sampah	pembaris
Norwegian	morgenkåpe	sofa	søppelbøtte	linjal
Polish	szlafrok	kanapa	kosz	linijka
Portuguese	robe	sofá	caixote de lixo	régua
Russian	халат	диван	мусорка	линейка
Serbian	bademantil	kauč	kanta	lenjir
Slovak	župan	gauč	kôš	pravitko
Spanish				
Peninsular	bata	sofá	papelera	regla
Rioplatense	bata	sillón	tacho	regla
Turkish	bornoz	koltuk	çöp kutusu	cetvel
Welsh	dressing gown	soffa	bin	pren mesur

of techno-socio-economic research, exemplified by exploration of esthetic complexity, also employs MultiPic, highlighting its versatility and growing acceptance in diverse scientific inquiries. At the same time, MultiPic remains an ongoing open-ended project, which invites interested parties to norm it in yet more languages.

Conclusion

MultiPic has represented a significant advancement in the field of cognitive science and psycholinguistics. It exemplifies the power of collaboration and open-source initiatives in bridging cultural and linguistic divides in research. As this tool continues to evolve, it promises to offer even deeper insights into the intricate relationship between cognition, language, and culture. The impact of MultiPic extends far beyond its current applications, paving the way for future research that is more inclusive, diverse, and representative of the global community.

References

- Brodeur, M. B., Dionne-Dostie, E., Montreuil, T., & Lepage, M. (2010). The Bank of standardized stimuli (BOSS), a new set of 480 normative photos of objects to be used as visual stimuli in cognitive research. *PLoS ONE*, 5(5), e10773. <https://doi.org/10.1371/journal.pone.0010773>.
- Dimitropoulou, M., Duñabeitia, J. A., Blitsas, P., & Carreiras, M. (2009). A standardized set of 260 pictures for Modern Greek: Norms for name agreement, age of acquisition and visual complexity. *Behavior Research Methods*, 41(2), 584–589.

- Duñabeitia, J. A., et al. (2018). MultiPic: A standardized set of 750 drawings with norms for six European languages. *Quarterly Journal of Experimental Psychology*, 71(4), 808–816.
- Duñabeitia, J. A., et al. (2022). The multilingual picture database. *Scientific Data*, 9.
- Kuhne, L. A., Ksiezarczyk, A. M., Braumann, K. M., Reer, R., Jacobs, T., Röder, B., et al. (2023). Cardiovascular exercise, learning, memory, and cytokines: Results of a ten-week randomized controlled training study in young adults. *Biological Psychology*, 176, 108466. <https://doi.org/10.1016/j.biopsycho.2022.108466>.
- Obidziński, M., & Nieznański, M. (2022). Context and target recollection for words and pictures in young adults with developmental dyslexia. *Frontiers in Psychology*, 13, 993384. <https://doi.org/10.3389/fpsyg.2022.993384>.
- Peristeri, E., Andreou, M., Ketseridou, S.-N., Machairas, I., Papadopoulou, V., Stravrovdi, A. S., et al. (2023). Animacy processing in Autism: Event-related potentials reflect social functioning skills. *Brain Sciences*, 13, 1656. <https://doi.org/10.3390/brainsci13121656>.
- Rossion, B., & Pourtois, G. (2004). Revisiting Snodgrass and Vanderwart's object pictorial set: The role of surface detail in basic-level object recognition. *Perception*, 33(2), 217–236. <https://doi.org/10.1068/p5117>.
- Santos, M. P. F., Justi, F. R. R., Buratto, L. G., Oliveira, B. S. F., & Jaeger, A. (2019). Brazilian norms for the Bank of standardized stimuli (BOSS). *PLoS ONE*, 14(11), e0224973. <https://doi.org/10.1371/journal.pone.0224973>.
- Snodgrass, J. G., & Vanderwart, M. (1980). A standardized set of 260 pictures: Norms for name agreement, image agreement, familiarity, and visual complexity. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 6, 174–215.
- Thompson, B., Roberts, S. G., & Lupyan, G. (2020). Cultural influences on word meanings revealed through large-scale semantic alignment. *Nature Human Behavior*, 4, 1029–1038. <https://doi.org/10.1038/s41562-020-0924-8>.
- Zhong, J., Huang, W., Kang, K., et al. (2024). Standardizing norms for 1286 colored pictures in Cantonese. *Behavior Research Methods*, 1–24. <https://doi.org/10.3758/s13428-024-02362-y>.